

CHANNING BLOEDEL

Brookings, SD 57006, (605) 591-2237, channingbloedel@gmail.com

SKILLS

Technical Skills: CAD (SolidWorks, Autodesk Inventor, Creo), Analysis (ANSYS, StarCCM+, ADAMS), MATLAB, Autodesk ReCap, Microsoft Office Proficiency, Basic Machining Proficiency, Basic Maintenance

EDUCATION

South Dakota State University, Brookings, SD

May 2028

- Pursuing Mechanical Engineering Bachelor of Science
- GPA 3.3/4.0
- Relevant coursework upon Summer 2027: Thermodynamics II, Heat Transfer, Automatic Controls, Aerodynamics, Design of Machine Elements, Engineering Materials, Engineering Design Methods, Advanced Engineering Math

WORK EXPERIENCE

Mechanical Design Intern – Aerofly

October 2025 – October 2026

- Digital twin modeling of 15+ electrical substations in Autodesk Inventor using Autodesk ReCap data
- Collaborated heavily with other employees to align designs across substation systems
- Led quality control efforts and created systematic processes to assist in efficient workflow

Lab Services Reliability Mechanical Engineering Student - Daktronics

April 2025 – September 2025

- Performed reliability engineering testing on power supply systems used in LED displays
- Conducted construction analysis to verify component integrity and design alignment
- Executed failure analysis investigations to determine root causes of malfunction in power systems
- Documented failure analysis findings and collaborated with engineering teams for product improvements
- Designed parts in Creo Design to further development of new testing methods

District Technology Assistant - Lead-Deadwood School

May 2024 – August 2024

- Assisted in the surplus process of over 200 pieces of hardware
- Imaged, and prepared technology for ~450 students

Freelance Maintenance - Miller Apartments

September 2016 – July 2024

- Ran repair and renovation jobs for a residential property, estimating costs on 50+ jobs

ACTIVITIES

- **NASA RASC-AL 2027 Project “TBD” – Team Leader** August 2026 – Present
 - **Systems engineering** competition to design a lunar transit pathway construction mission
 - Leading a team of 15 to start, grow, and accomplish in the aerospace field
- **NASA RASC-AL 2026 Project “SELENE” – Design Lead** March 2026 – June 2026
 - **Systems engineering** competition to design a sample storage mission to the moon
 - **Designed** a refrigerated and ambient storage container to hold lunar surface core samples
 - Selected as **Best in Theme (Finalist)**
- **NASA RASC-AL 2026 Project “PROMETHEUS” - Team Leader** September 2025 – March 2026
 - **Systems engineering** competition to design an architecture to power the moon
- **NASA RASC-AL 2025 Project “NOVA” - Mobility Sub-Team Leader** September 2024 – June 2025
 - **Systems engineering** competition to **design** a lunar **maintenance** and **servicing rover**
 - **Lead collaboration** within a small team to design an effective lunar rover mobility system
 - **Designed and FEA'd** rover mobility system using SolidWorks, ANSYS, and ADAMS View, and **Prototyping**
 - Awarded **Best Prototype (Finalist)**
- **NASA X-Hab Project** April 2025 – December 2025
 - Assisted on proposal team to obtain this project for South Dakota State University's senior design class
 - Participated in project as a helping hand, completing time sensitive objectives
- **American Society of Mechanical Engineers (ASME)- Member** September 2024 – Present
- **Oasis Leadership Team** March 2025 – Present